MAYER-VIETORIS DISTINGUISHED TRIANGLE FOR THE DERIVED PUSHFORWARD ALONG THE INCLUSIONS OF TWO OPEN SUBSPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof defers the exactness of the sequence of sections to the proof of the Mayer-Vietoris lemma, which is a navigation pointer; the termwise statement that makes the appeal applicable is written out inside the counted span.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-exact-sequence-j-star is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- K-injective resolutions with injective terms exist. Injectives, Theorem injectives-theorem-K-injective-embedding-grothendieck, invoked at proof lines 8136-8137. Supplies the representative by which the derived pushforwards are computed.
- Restriction to an open subspace preserves K-injectivity. Same chapter, Lemma lemma-restrict-K-injective-to-open, invoked at proof lines 8138-8139. Supplies that the restricted complexes compute the derived pushforwards on U, V and the intersection.
- Distinguished triangle attached to a short exact sequence of complexes. Derived Categories, Section derived-section-canonical-delta-functor and Lemma derived-lemma-derived-canonical-delta-functor, invoked at proof lines 8141-8145. Supplies the triangle once the short exact sequence of complexes is in hand.
- The termwise exact sequence of sections over W, W cap U, W cap V and W cap U cap V. Same chapter, Lemma lemma-mayer-vietoris, whose proof is cited for this exactness at proof lines 8154-8163. Supplies the exactness of the short exact sequence of complexes of sections.

Source reference. Original lemma 0152: sources/stacks-chow/tags.txt, source lines 1493–1493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original theorem 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EB: sources/stacks-cohomology/pinned-original-source.tex, source lines 767–767. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08BS: sources/stacks-cohomology/pinned-original-source.tex, source lines 7840–7840. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $X = U \cup V$ be the union of two open subspaces. For any object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*

$$E \to Rj_{U,*}E|_U \oplus Rj_{V,*}E|_V \to Rj_{U\cap V,*}E|_{U\cap V} \to E[1]$$

in $D(\mathcal{O}_X)$.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E whose terms $\mathcal{I}^n$ are injective objects of $Mod(\mathcal{O}_X)$, see Injectives, Theorem 079P. We have seen that $\mathcal{I}^\bullet|U$ is a K-injective complex as well (Lemma 08BS). Hence $Rj_{U,*}E|_U$ is represented by $j_{U,*}\mathcal{I}^\bullet|_U$. Similarly for V and $U \cap V$. Hence the distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes

$$0 \to \mathcal{I}^\bullet \to j_{U,*}\mathcal{I}^\bullet|_U \oplus j_{V,*}\mathcal{I}^\bullet|_V \to j_{U\cap V,*}\mathcal{I}^\bullet|_{U\cap V} \to 0.$$

This sequence is exact because for any $W \subset X$ open and any n the sequence

$$0 \to \mathcal{I}^n(W) \to \mathcal{I}^n(W \cap U) \oplus \mathcal{I}^n(W \cap V) \to \mathcal{I}^n(W \cap U \cap V) \to 0$$

is exact (see proof of Lemma 01EB). □